Jinhyeok Park

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OBJECTIVE

Ph.D. student in ECE developing resource-efficient intelligent sensing systems, spanning mmWave/RF hardware, radar signal processing, state-space models, and real-time embedded ML.

EDUCATION

Ph.D., Electrical and Computer Engineering, Georgia Institute of Technology, USA *Aug 2024 – Present*
GPA: 3.85/4.0 | Expected Graduation: Aug 2028 | Advisor: Prof. Saibal Mukhopadhyay

M.S., Electrical Engineering, Korea Advanced Institute of Science and Technology (KAIST), South Korea *Mar 2021 – Feb 2023*

B.S., Electrical Engineering, Sungkyunkwan University, South Korea *Mar 2014 – Feb 2021*

WORK EXPERIENCE

RF Engineer, Samsung Electronics –South Korea | Advisor: Dr. Hyun-chul Park *Mar 2023 – Apr 2024*

- Developed 60 GHz FMCW Radar built-in-self-test modules for mobile sensing applications.
- Designed and validated 28/39 GHz phased-array RF transceiver modules for commercial 5G products, covering RF circuit design, tuning, and module-level characterization.

• RESEARCH EXPERIENCE

Graduate Research Assistant (PhD), Georgia Tech, GREEN Lab *Aug 2024 – Present*

- Developed real-time adaptive RF front-end and baseband control on TI mmWave hardware.
- Lead research on intelligent sensing: physics-informed state-space models for NLOS imaging, tracking, and filtering, from algorithm design to evaluation on real mmWave data.
- Develop statistically guaranteed cooperative perception using conformal prediction for risk-controlled sensor fusion.
- Co-developed efficient SSMs, including readout-aware token pruning and dynamics-grounded latent-state communication.

Graduate Research Assistant (MS), KAIST, WEIS Lab *Mar 2021 – Feb 2023*

- Designed a 23–35 GHz wideband bidirectional VGA and 24-30 GHz power amplifier for 5G beamforming ICs.

PUBLICATIONS

[1] **J. Park** et al., "Anytime NLOS Radar Imaging with Complex-Valued Selective State-Space Models" *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2027.

[2] **J. Park** et al., "Range-Adaptive FMCW Radar: Real-Time Adaptation of RF and Baseband for Robust and Energy-Efficient Sensing," *IEEE/MTT-S International Microwave Symposium (IMS)*, 2026.

[3] **J. Park** et al., "Parallel Kalman Filtering with Physics-Informed Selective State-Space Models for Robust Radar Sensing," *IEEE/MTT-S International Microwave Symposium (IMS)*, 2026.

[4] **J. Park**, S. Hong. "Wideband Bidirectional Variable Gain Amplifier for 5G Communication," *IEEE Microwave and Wireless Technology Letters*, 2023.

[5] S. Sharma, **J. Park** et al., "LUGA: Lightweight Uncertainty-Guided Sensing Resolution Adaptation for Energy Efficient Radar Processing," *IEEE SENSORS 2025*.

[6] **J. Park** et al., "Ego-Sovereign Fusion: Risk-Controlled Preservation for Cooperative Perception" (ICRA 2027 under review)

[7] H. Cho, **J. Park** et al, "State as Messages: Latent Communication for State Space Models" (ICLR 2027 under review)

[8] A. Padhy, **J. Park** et al, "ReBound: Readout-Aware Boundary Pruning for Efficient Mamba Prefill" (ICLR 2027 under review)

TECHNICAL SKILLS

ML / Sequence modeling: state-space models, physics-informed learning, Kalman filtering, conformal prediction
Signal Processing: FFT/Range-Doppler, CFAR, Digital beamforming (Capon), SAR/coherent integration, waveform design
Hardware & Embedded: TI mmWave sensor (AWR1843), FPGA acceleration (RFSoC), real-time DSP, Cadence Virtuoso, ADS
Programming: Python, PyTorch, MATLAB, C/C++

AWARDS AND HONORS

Best Poster Award, Cognisense Annual Review, SRC, USA *Oct 2025*
Georgia Tech ECE Fellowship *Aug 2024 – May 2025*
Samsung Semiconductor Scholarship *Mar 2021 – Feb 2023*